

Introduction

Algebra is used to create mathematical models of real-world situations and to handle problems that we can't solve using just arithmetic. Instead of using numbers or words, algebra uses symbols to make statements about things. In algebra, we often use letters to represent numbers.

8.1 ALGEBRAIC EXPRESSIONS

We have already learnt in previous class that an algebraic expression consists of terms which are connected by the operation of addition (+) and subtraction (-).

For examples: $4x + 5y + 3$, $6a + 5b - 4$, $5m \div 6 \times 2p \times q$

8.1.1 Define a constant as a symbol having a fixed numerical value.

A symbol having a fixed numerical value is called a constant.

For example, 3, 8, 12, ... etc are all constants.

8.1.2 Recall variable as a quantity which can take various numerical values.**(a) Variables**

These are represented by symbols or letters. For example, x is the number of rupees in the pocket of a student. It can take values in rupees 5, 10, 35, 50 etc.

(b) Coefficients

Coefficient is the number multiplied by a variable or variables.

In $3x$, the coefficient is 3.

In $5yz$, the coefficient is 5.

8.1.3 Recall literal as an unknown number represented by an alphabets

The letters or alphabets that we use to represent unknowns are called literal numbers. For example, area of square can be calculated by multiplying its side by side i.e Area of square $A = \text{side} \times \text{side}$ or $A = s \times s$.

Similarly area of a rectangle can be calculated by multiplying length and breadth i.e

Area = Length $\times$ Breadth or $A = l \times b$. In the above example, s is used for side, l is used for length and b is used for breadth.

Thus s , l and b etc represent the unknown quantities.

8.1.4 Recall algebraic expression as a combination of constants and variables connected by the signs of fundamental operations.

We know that an algebraic expression is the combination of variables and constants connected by the signs of fundamental operations i.e. $+$, $-$, $\div$, $\times$.

For example,

(a) $2x + 3y$ (b) $2x^2 + 5y^2 + 3y$ (c) $5a \div b + 3c$ and (d) $4m \times n - 2q$ are some algebraic expressions.

8.1.5 Define polynomial as an algebraic expression in which the powers of variables are all whole numbers.

In algebra polynomial represents an algebraic expression containing a single term as well as two or more than two terms.

For a polynomial, the exponents of the variables must be the whole numbers. For example 6 , 7 , $2x$, $4x$, $x^2 + 5$, $y^2 + 3$, $3x + 3x^2 + x + 2$ are polynomials but $y + \frac{1}{y}$, $x^2 + \frac{1}{x}$, $x^3 + y^{\frac{3}{2}} + 3$ etc, are not polynomials because their exponents of variables are not whole numbers.

Thus an algebraic expression in which the exponent is a whole number is called a polynomial.

Therefore we define a Polynomial as:

An algebraic expression consisting of one or more terms, each of which exponent (index) of variable is whole number is called polynomial.

8.1.6 Identify a monomial, a binomial and a trinomial as a polynomial having one term, two terms and three terms respectively.**1. Monomial:**

An expression having one term is called monomial or simple expression i.e. 2 , $3x$, $4ab$ etc.

2. Binomial:

An expression having two terms is called binomial i.e. $x + y$ and $3a - bx^2$, $x^2 + x$ etc.

3. Trinomial:

An expression having three terms is called trinomial i.e. $a^3 + 2a - b$, $2x + 3g + 4z$ and $x^3 + x^2 + x$.

Examples of polynomials:

Polynomial	Number of terms	Some examples
Monomial	1	2, x , $5x^3$
Binomial	2	$2x + 5$, $x^2 - x$, $x - 5$
Trinomial	3	$x^2 + 5x + 6$, $x^5 - 3x + 8$

Activity: Identify and tick (✓) the given polynomial is a monomial, a binomial, or trinomial.

S. No.	Polynomial	Monomial	Binomial	Trinomial
1)	$3.4 + 3.4x$			
2)	$z^2 + 5z + 6$			
3)	-8			
4)	$2c + 5b + 6$			
5)	$14 + x$			
6)	$5x - 2$			
7)	$4b - 2a$			
8)	$f^2 + 5f + 6$			

8.2 OPERATIONS WITH POLYNOMIALS

8.2.1 Add two or more polynomials:

Addition of polynomials:

Let us solve some examples to understand the addition.

Example 1: Add $7a$ and $3a$

Solution: $7a + 3a = (7 + 3)a$, [Distributive Property of Multiplication]
 $= 10a$

Example 2: Add $6ab$, $5ab$ and $-7ab$

Solution: Horizontal Method

$$6ab + 5ab - 7ab = (6 + 5 - 7) ab = (11 - 7) ab = 4ab$$

Vertical Method

$$\begin{array}{r} 6ab \\ 5ab \\ -7ab \\ \hline 4ab \end{array}$$

In vertical method, similar terms are written one below the other and their coefficients are added and simplified.

Example 3:

Add $3a + 2b$ and $2a + 5b$

Solution: Horizontal Method

$$\begin{aligned} (3a + 2b) + (2a + 5b) \\ = 3a + 2b + 2a + 5b \\ = 3a + 2a + 2b + 5b \\ = (3 + 2)a + (2 + 5)b \\ = 5a + 7b \end{aligned}$$

Vertical Method,

$$\begin{array}{r} 3a + 2b \\ 2a + 5b \\ \hline 5a + 7b \end{array}$$

Example 4:

Add $6a^2b - 3ab^2$ and $4a^2b + 3ab^2$

Solution: Horizontal Method

$$\begin{aligned} (6a^2b - 3ab^2) + (4a^2b + 3ab^2) \\ = (6a^2b + 4a^2b) + (-3ab^2 + 3ab^2) \\ = 10a^2b + 0 \\ = 10a^2b \end{aligned}$$

Vertical Method,

$$\begin{array}{r} 6a^2b - 3ab^2 \\ 4a^2b + 3ab^2 \\ \hline 10a^2b + 0 \end{array} \quad \text{or } 10a^2b$$

In vertical method the given expressions are written one below the other in such a way that the similar terms are in one and the same column. Then these similar terms are added. This gives the addition of the given expressions.

If the sum of the coefficients is zero, then the whole term becomes zero.

EXERCISE 8.1

A. Add the following:

- (1) $5a, 8a$ (2) a, a (3) $-12x, -2x$
 (4) $9ab, -7ab$ (5) $8y, 3y, -4y$ (6) $6x^2 + 2x + 4, 3x^2 + 5x$

B. Find the sum of:

- (1) $2a + 3b, 6a + 5b$ (2) $3x - 4y, 4x + 6y$
 (3) $2a - 3b, 4a + 5b, 5a - 8b$ (4) $ab + 4cd + 6ad, 8ab + 5cd + 6ad$
 (5) $4.5a + 5.7b + 3.2c, 3.5a + 1.3b + 2.8c$
 (6) $1.7a + 3.2b + 4.9c, 4.3a + 3.2b + 3.1c, 6.0a + 9.6b + 3.0c$
 (7) $6a + 11b + 10c, 13a + 9b + 5c, 8a + 7b + 12c$
 (8) $4a + 6b + 5c, 7a + 5b + 5c, 3a + 7b + 7c$
 (9) $2a + 3b + 4c, 6a + 8b + 4c, 3a + 2b + 3c$
 (10) $14e - 15f + 10g, 13e + 14f - g, -5e + 3g$

8.2.2 Subtract a polynomial from another polynomial

We know that the operation of subtraction is the inverse operation of addition. According to the rules of directed numbers:

$$5 - 2 = 5 + (-2)$$

and $a - b = a + (-b)$

This means that to subtract b from a , we add the additive inverse of b , i.e. $-b$ to a .

Thus, to subtract one expression from the other, we change the signs of the terms of the expression to be subtracted and then add them.

Example 1. Subtract **$2a + 5b$ from $3a + 8b$** **Vertical method**

Solution:

$$\begin{array}{r}
 3a + 8b \\
 - 2a + 5b \\
 \hline
 1a + 3b
 \end{array}
 \quad \text{or} \quad a + 3b$$

Horizontal method

$$\begin{aligned}
 & (3a + 8b) - (2a + 5b) \\
 &= 3a + 8b - 2a - 5b \\
 &= 3a - 2a + 8b - 5b \\
 &= (3 - 2)a + (8 - 5)b \\
 &= 1a + 3b \\
 &= a + 3b
 \end{aligned}$$

Example 2: Subtract $2x + 3y - 4z$ from $3x - 2y + 5z$

Solution:

$$\begin{array}{r} 3x - 2y + 5z \\ \pm 2x \pm 3y \mp 4z \\ \hline x - 5y + 9z \end{array}$$

Example 3: Subtract $4x - y + 2$ from the sum of $3x + 8y + 1$ and $2x + 3y - 6$.

Solution:

$$\begin{array}{r} 3x + 8y + 1 \\ 2x + 3y - 6 \\ \hline \text{Sum: } 5x + 11y - 5 \\ \pm 4x \quad \mp y \pm 2 \\ \hline \text{Difference: } x + 12y - 7 \end{array}$$

EXERCISE 8.2

Subtract the second expression from the first expression.

- (1) $8a, 5a$
- (2) $-7ab, 12ab$
- (3) $4a, -8a$
- (4) $6a + 7b, 2a + 3b$
- (5) $9a + 11b + 15c, 5a + 3b - 6c$
- (6) $9c + 8d + 6e, 3e - 5c + 4d$
- (7) $18x + 16y + 12z, 10x + 9y + 8z$
- (8) $14a + 15b + 16c, 8a + 10b - 2c$
- (9) $3x^2 + 25ab + 21bc, 4x^2 + 18ab - 6bc$
- (10) $5a + 4ab + 2b, 4a + 3ab + b$
- (11) Subtract $10a + 10b + 17c$ from the sum of $13a + 8b + 12c$ and $9a + 3b + 5c$
- (12) What should be added to $2x + 5y$ to get $6x + 8y$?

8.2.3 Find the product of:

- monomial with monomial
- monomial with binomial and trinomial
- binomial with binomial trinomial.

I. Multiplication of monomial with monomial

While multiplying monomials, the coefficients and variables in each of the corresponding terms are multiplied separately.

Example 1: Find the product of $2a$ and $3a^2$

Solution: $2a \times 3a^2$
 $= 2 \times 3 \times a \times a^2 = 6a^{1+2} = 6a^3$

Example 2: Find the product of $3ay^2$ and $7a^2y^3$

Solution: $3ay^2 \times 7a^2y^3 = 3 \times 7 \times a \times a^2 \times y^2 \times y^3$
 $= 21 \times a^{1+2} \times y^{2+3} = 21a^3y^5$

II. Multiplication of monomial with binomial.

Example 3: Multiply $(a + b)$ by 3

Solution: Horizontal Method

$$3 \times (a + b) = 3 \times a + 3 \times b = 3a + 3b$$

Vertical Method

$$\begin{array}{r} a + b \\ \times \quad 3 \\ \hline 3a + 3b \end{array}$$

Example 4: Multiply $2a + 3b$ by a

Vertical Method.

Solution: $\begin{array}{r} 2a + 3b \\ \times \quad a \\ \hline 2a^2 + 3ab \end{array}$

Horizontal Method.

$$a \times (2a + 3b) = 2a^2 + 3ab.$$

III. Multiplication of monomial with trinomial

Example 5. Multiply $(x + y + z)$ by $4x$

Solution:

Vertical Method.

$$\begin{array}{r} x + y + z \\ \times 4x \\ \hline 4x^2 + 4xy + 4xz \end{array}$$

Horizontal Method.

$$\begin{aligned} 4x(x + y + z) \\ = 4x^2 + 4xy + 4xz \end{aligned}$$

IV. Multiplication of binomial with binomial.

Example 6: Multiply $3e + 5f$ by $4e + 2f$

Solution:

Vertical Method.

$$(3e + 5f) \times (4e + 2f)$$

$$\begin{array}{r} 3e + 5f \\ \times 4e + 2f \\ \hline 12e^2 + 20ef \\ + \quad 6ef + 10f^2 \\ \hline 12e^2 + 26ef + 10f^2 \end{array}$$

V. Multiplication of binomial with trinomial.**Example 7: Multiply $2x^2 - 3xy + 4$ by $x + y$** **Solution: Vertical Method**

$$\begin{array}{r}
 2x^2 - 3xy + 4 \\
 \times \quad \quad x + y \\
 \hline
 2x^3 - 3x^2y + 4x \\
 + 2x^2y \quad - 3xy^2 + 4y \\
 \hline
 2x^3 - x^2y + 4x - 3xy^2 + 4y
 \end{array}$$

Horizontal Method

$$\begin{aligned}
 (x + y)(2x^2 - 3xy + 4) &= x(2x^2 - 3xy + 4) + y(2x^2 - 3xy + 4) \\
 &= 2x^3 - 3x^2y + 4x + 2x^2y - 3xy^2 + 4y \\
 &= 2x^3 - 3x^2y + 2x^2y + 4x - 3xy^2 + 4y \\
 &= 2x^3 - x^2y + 4x - 3xy^2 + 4y
 \end{aligned}$$

EXERCISE 8.3**I. Find the product of the following expressions: (Monomial with monomial)**

- (1) $15a, 8a$ (2) $9a, 6b$ (3) $10a^2, 2a^5$ (4) $4c^5, 8cd^2$
 (5) $6a^2b, -5b^2$ (6) $-3a, 2b, -3$ (7) $4m^2, -5m^3, 3m^4$ (8) $-6xy, 3x^2y^2, 2xy$

II. Find the product of following expressions: (Monomial with binomial).

- (1) $a + b, 9$ (2) $x - y, -4y$ (3) $2m^2 - 3n^3, -5mn$
 (4) $3xy + 2xy^2, -6xy^2$ (5) $6lm - 4mn, 3l^2m^2n$ (6) $-5bc + 3cd, -2bc^2d$

III. Find the product of the following expressions: (Binomial with binomial).

- (1) $8c + 12d, -3c + 2d$ (2) $a + b, a - b$ (3) $a^2 + b^2, a^2 - b^2$
 (4) $3x - 2y, 2x + 5y$ (5) $2lm + 3mn, -2m + 3n$ (6) $8pq^2 - 3p^2q, 2pq$

IV. Find the product of the following expressions (Monomial with trinomial).

- (1) $2a - 3b + 4c$, $2abc$ (2) $a^2 + b^2 - c^2$, ab (3) $m^2n + mn^2 - m^2n^2$, $-mn$
 (4) $x^2 - 6xy + 9y^2$, $-3xy$ (5) $2p^2 - 3pq + q^2$, $4pqr$ (6) $xyz + x^2y^2 - y^2z^2$, xyz

V. Find the product of the following expression (Binomial with trinomial).

- (1) $1 + 2x - 3x^2$, $1 - x$ (2) $8c + 12d - 6e$, $c^2 - d^2$ (3) $a^2 - ab - b^2$, $a - b$
 (4) $x^2 - 5x + 4$, $3x - 4$ (5) $a^2 - 2a^2b + b^2$, $a^2 - b^2$ (6) $3p^2q + 2pq^2 - q^2r$, $p^2 - q^2$

8.2.4 Simplify algebraic expressions involving addition, subtraction and multiplication.

We have already learnt addition, subtraction and multiplication of algebraic expressions. We will learn to simplify the expressions involving these operations through following examples.

Example. Simplify: (i) $3x(x + 2y) + 4x^2 - 7xy$

(ii) $2x + \{ (3x + y)(2x - y) - (6x^2 - 2x) \}$

Solution:

$$\begin{aligned} \text{(i)} \quad & 3x(x + 2y) + 4x^2 - 7xy \\ &= 3x^2 + 6xy + 4x^2 - 7xy \\ &= 3x^2 + 4x^2 + 6xy - 7xy \\ &= 7x^2 - xy \end{aligned}$$

Solution:

$$\begin{aligned} \text{(ii)} \quad & 2x + \{ (3x + y)(2x - y) - (6x^2 - 2x) \} \\ &= 2x + \{ (3x + y)(2x - y) - (6x^2 - 2x) \} \\ &= 2x + \{ 3x(2x - y) + y(2x - y) - 6x^2 + 2x \} \\ &= 2x + \{ 6x^2 - 3xy + 2xy - y^2 - 6x^2 + 2x \} \\ &= 2x + \{ -xy - y^2 + 2x \} \\ &= 2x - xy - y^2 + 2x \\ &= -xy - y^2 \end{aligned}$$

8.3 Algebraic Identities

Recognize and verify the algebraic identities.

An algebraic identity is a simplified form consisting of the algebraic term which provide us with a rule to solve any algebraic expression.

Identities are some such results which are repeatedly used in algebra. We can easily find the product of expressions by using these identities.

8.3.1 Verify: $(x + a)(x + b) = x^2 + (a + b)x + ab$

$$\begin{aligned} \text{Verification: } (x + a)(x + b) &= x^2 + (a + b)x + ab \\ \text{LHS} = (x + a)(x + b) &= x \times (x + a) + b \times (x + a) \\ &= x^2 + ax + bx + ab \\ &= x^2 + (a + b)x + ab = \text{RHS} \end{aligned}$$

Thus LHS = RHS.

Hence verified.

By this identity we can multiply two linear expressions quickly.

Hence, $(x + a)(x + b) = x^2 + (a + b)x + ab$

We can also verify it by using geometrical shape.

	10	2	
10	100	20	
4	40	8	

The expression can also be worked out as follows, without drawing a graph

$$\begin{aligned} (10 + 4)(10 + 2) &= 10^2 + (4 + 2) \times 10 + 4 \times 2 \\ &= 100 + 6 \times 10 + 8 \\ &= 100 + 60 + 8 \\ &= 168 \end{aligned}$$

The expression $(x + 4)(x + 2)$ is expanded as shown:

	x	2	
x	x^2	$2x$	
4	$4x$	8	

$$\begin{aligned} (x + 4)(x + 2) &= x^2 + 4x + 2x + 8 \\ &= x^2 + 6x + 8 \end{aligned}$$

Similarly, we can verify:

$$(i) \quad (x + a)(x - b) = x^2 + (a - b)x - ab$$

$$(ii) \quad (x - a)(x + b) = x^2 - (a - b)x - ab$$

$$(iii) \quad (x - a)(x - b) = x^2 - (a + b)x + ab$$

Example 1: Verify: $(x + 2)(x + 3) = x^2 + 5x + 6$

Verification:

$$\begin{aligned} \text{LHS: } (x + 2)(x + 3) &= x^2 + (2 + 3)x + (2) \times (3) \\ &= x^2 + 5x + 6 = \text{RHS} \end{aligned}$$

Thus LHS = RHS Hence verified

Example 2: Verify: $(x - 2)(x + 3) = x^2 + x - 6$

Verification:

$$\begin{aligned} \text{LHS: } (x - 2)(x + 3) &= x^2 + (-2 + 3)x + (-2) \times (3) \\ &= x^2 + x - 6 = \text{RHS} \end{aligned}$$

Thus LHS = RHS Hence verified

Example 3: Verify: $(x - 2)(x - 3) = x^2 - 5x + 6$

Verification: Using identity:

$$\begin{aligned} \text{LHS: } (x - 2)(x - 3) &= x^2 + (-2 - 3)x + (-2) \times (-3) \\ &= x^2 - 5x + 6 = \text{RHS} \end{aligned}$$

Thus LHS = RHS Hence verified

EXERCISE 8.4

I. Simplify:

$$(1) \quad 5a - 2a(3 + 4b)$$

$$(2) \quad 7m - 4m(5 - 6n)$$

$$(3) \quad 6x\{2x(3x^2 - 4) - (6x^2 + 8y)\} \\ \quad \quad \quad 2y^2\}$$

$$(4) \quad 3x - \{(2x + y)(2x - y) - (4x^2 +$$

II. Verify the following.

$$(1) \quad (x + 3)(x + 4) = x^2 + 7x + 12$$

$$(2) \quad (x + 5)(x + 6) = x^2 + 11x + 30$$

$$(3) \quad (x + 2)(x - 4) = x^2 - 2x - 8$$

$$(4) \quad (x - 3)(x + 4) = x^2 + x - 12$$

$$(5) \quad (x - 5)(x - 4) = x^2 - 9x + 20$$

$$(6) \quad (x - 7)(x + 8) = x^2 + x - 56$$

$$(7) \quad (x - 9)(x + 2) = x^2 - 7x - 18$$

$$(8) \quad (x - 1)(x + 9) = x^2 + 8x - 9$$

$$(9) \quad (x - 2)(x + 5) = x^2 + 3x - 10$$

8.3.2 Verify $(a + b)^2 = (a + b)(a + b) = a^2 + 2ab + b^2$

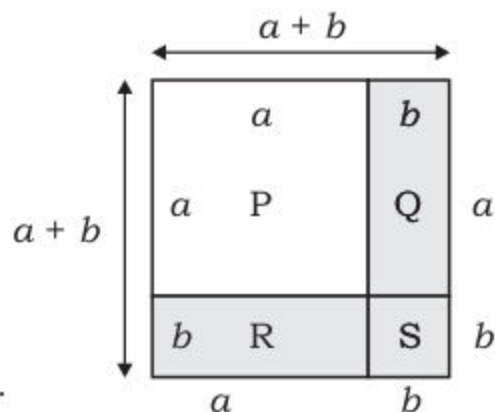
Geometrical verification: Consider the area of a square whose side is $(a + b)$

The area of the square whose sides are $(a + b)$ is equal to the sum of the areas of P, Q, R and S.

$$\text{LHS} = (a + b)^2$$

$$\begin{aligned} &= (a + b)(a + b) = a^2 + ab + ab + b^2 \\ &= a^2 + 2ab + b^2 \\ &= a^2 + 2ab + b^2 = \text{RHS} \end{aligned}$$

Thus LHS = RHS Hence verified.



Algebraic verification:

$$(a + b)^2 = a^2 + 2ab + b^2$$

This identity also can also be verified as:

Vertical Method

$$\begin{array}{r} \text{LHS} = (a + b)^2 = (a + b)(a + b) \\ \begin{array}{r} a + b \\ \times a + b \\ \hline a^2 + ab \\ + ab + b^2 \\ \hline a^2 + 2ab + b^2 \end{array} \end{array}$$

(Multiply $a + b$ by a)

(Multiply $a + b$ by b)

$$a^2 + 2ab + b^2 = \text{RHS}$$

Horizontal Method

$$\begin{aligned} \text{LHS} &= (a + b)^2 = (a + b)(a + b) \\ &= a(a + b) + b(a + b) \\ &= a^2 + ab + ba + b^2 \text{ (because } ab = ba) \\ &= a^2 + 2ab + b^2 = \text{RHS} \end{aligned}$$

Thus LHS = RHS

Hence verified

$$(a + b)^2 = a^2 + 2ab + b^2$$

This identity can be described in words.

'Square of the sum of two terms is always equal to the square of the first term plus twice the product of first and second terms plus the square of the second term.'

Example 1: Verify: $(a^2 + 5)^2 = a^2 + 2ab + b^2$

Verification:

$$\begin{aligned} (a + 5)^2 &= (a + 5)(a + 5) \\ &= a + 2 \times a \times 5 + 5^2 \\ &= a^2 + 10a + 25 = \text{RHS} \end{aligned}$$

Thus LHS = RHS

Hence verified

Example 2: Verify $(3a + 2b)^2 = 9a^2 + 12ab + 4b^2$

Verification: LHS = $(3a + 2b)^2 = (3a + 2b)(3a + 2b)$
 $= (3a)^2 + 2(3a)(2b) + (2b)^2$
 $= 9a^2 + 12ab + 4b^2 = \text{RHS}$

Thus LHS = RHS

Hence verified

EXERCISE 8.5

I. Verify the following:

(1) $(a + 7)^2 = a^2 + 14a + 49$

(2) $(a + 11)^2 = a^2 + 22a + 121$

(3) $(2x + 6)^2 = 4x^2 + 24x + 36$

(4) $(3a + 5)^2 = 9a^2 + 30a + 25$

(5) $(7x + 3)^2 = 49x^2 + 42x + 9$

(6) $(6x + 9)^2 = 36x^2 + 108x + 81$

(7) $(5a + 4)^2 = 25a^2 + 40a + 16$

(8) $(9x + 10)^2 = 81x^2 + 180x + 100$

(9) $(5x + 11)^2 = 25x^2 + 110x + 121$

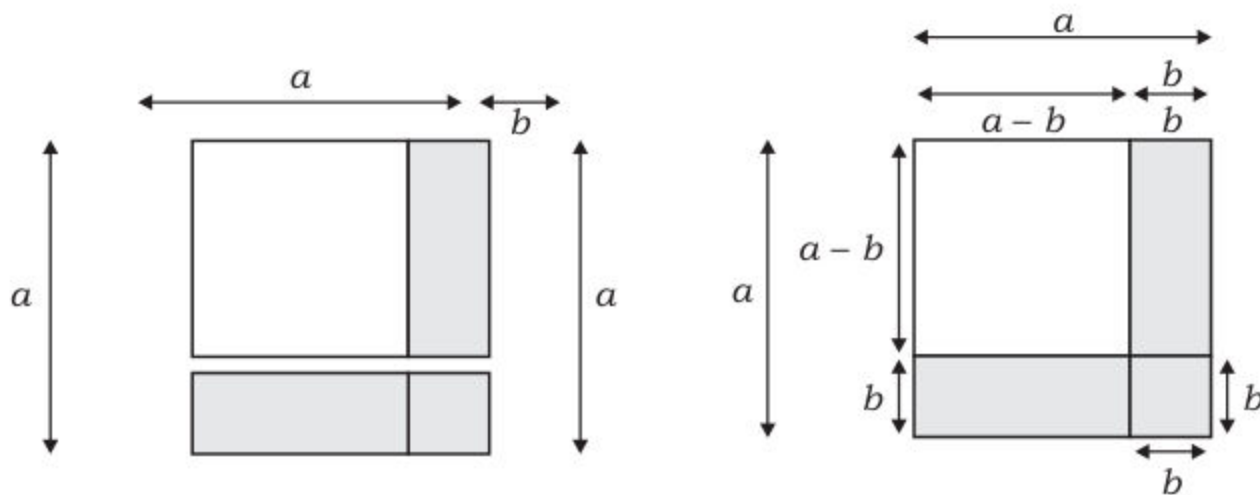
(10) $(6x + 7)^2 = 36x^2 + 84x + 49$

8.3.3 Verify $(a - b)^2 = (a - b)(a - b) = a^2 - 2ab + b^2$

Geometrical verification:

Consider the area of a square whose sides are a and the area of two rectangles whose length and width are a and b respectively.

The two rectangles are then placed on the square like this:



The area of the unshaded square is equal to $(a - b)(a - b)$

$$\begin{aligned} \text{LHS} &= (a - b)^2 = (a - b)(a - b) = \text{Area of original square of sides } a \\ &\quad - \text{Area of two rectangles of sides } a \text{ and } b \text{ (because the area of the square is subtracted twice).} \\ &\quad + \text{Area of small square of sides } b \\ &= a^2 - ab - ab + b^2 \\ &= a^2 - 2ab + b^2 = \text{RHS} \end{aligned}$$

Thus LHS = RHS

Hence verified: $(a - b)^2 = a^2 - 2ab + b^2$

Algebraic verification: We can also verify it as under:

Vertical Method

$$\begin{array}{r} \text{LHS: } (a - b)^2 = (a - b)(a - b) \\ \text{or} \quad \begin{array}{r} a - b \\ \times \quad a - b \\ \hline a^2 - ab \\ \quad - ab + b^2 \\ \hline a^2 - 2ab + b^2 \end{array} \end{array}$$

(Multiply $a - b$ by a)
(Multiply $a - b$ by b)

= RHS

Thus LHS = RHS

Horizontal Method

$$\begin{aligned} \text{LHS: } (a - b)^2 &= (a - b)(a - b) \\ &= a(a - b) - b(a - b) \\ &= a^2 - ab - ba + b^2 \text{ (because } ab = ba) \\ &= a^2 - 2ab + b^2 = \text{RHS} \end{aligned}$$

Hence verified

The identity can be described in words as:

‘Square of the difference of two terms is always equal to the square of the first term minus twice the product of first and second terms plus the square of the second term.’

Example 1: Verify $(a - 4)^2 = a^2 - 8a + 16$

Verification: LHS = $(a - 4)^2 = (a - 4)(a - 4)$

$$= a^2 - 2 \times a \times 4 + (4)^2$$

$$= a^2 - 8a + 16 = \text{RHS}$$

Thus LHS = RHS

Hence verified.

Example 2: Verified $\left(\frac{4}{5}x - \frac{1}{3}y\right)^2 = \frac{16}{25}x^2 - \frac{8}{15}xy + \frac{1}{9}y^2$

Verification: LHS = $\left(\frac{4}{5}x - \frac{1}{3}y\right)^2 = \left(\frac{4}{5}x - \frac{1}{3}y\right)\left(\frac{4}{5}x - \frac{1}{3}y\right)$

$$= \left(\frac{4}{5}x\right)^2 - 2 \times \left(\frac{4}{5}x\right) \times \left(\frac{1}{3}y\right) + \left(\frac{1}{3}y\right)^2$$

$$= \frac{16}{25}x^2 - \frac{8}{15}xy + \frac{1}{9}y^2 = \text{RHS}$$

Thus LHS = RHS

Hence verified

EXERCISE 8.6

Verify the following:

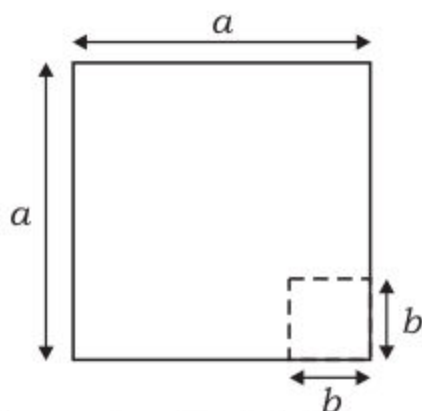
- | | |
|--------------------------------------|---|
| (1) $(a - 3)^2 = a^2 - 6a + 9$ | (2) $(a - 7)^2 = a^2 - 14a + 49$ |
| (3) $(2a - 5)^2 = 4a^2 - 20a + 25$ | (4) $(6x - 8)^2 = 36x^2 - 96x + 64$ |
| (5) $(4a - 7)^2 = 16a^2 - 56a + 49$ | (6) $(3a - 8)^2 = 9a^2 - 48a + 64$ |
| (7) $(7a - 5)^2 = 49a^2 - 70a + 25$ | (8) $(9y - 10)^2 = 81y^2 - 180y + 100$ |
| (9) $(8a - 9)^2 = 64a^2 - 144a + 81$ | (10) $(9a - 11)^2 = 81a^2 - 198a + 121$ |

8.3.4 Verify: $a^2 - b^2 = (a - b)(a + b)$

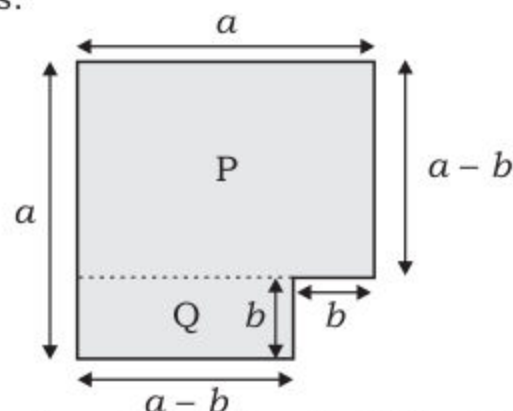
Geometrical verification:

Consider a square of sides a .

A small square of sides b is to be cut out on the square of sides a .



After the square of sides b is cut out, we get a figure which is comprised of rectangle P and a rectangle Q like this:



The rectangle Q is then cut out and placed next to rectangle P like this:

Here the area of the shaded rectangle is: $(a + b) + (a - b)a + b$

RHS = $(a + b)(a - b)$ Area of rectangle

P + Area of rectangle Q

$$= a(a - b) + b(a - b)$$

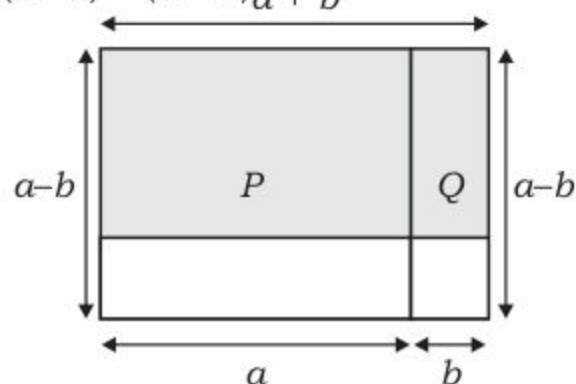
$$= a^2 - ab + ab - b^2$$

$$= a^2 - b^2 = \text{LHS}$$

Thus RHS = LHS

Hence verified

Therefore $(a + b)(a - b) = a^2 - b^2$



Given identity: $(a + b)(a - b) = a^2 - b^2$

Algebraic verification:

Multiplying $(a + b)$ and $(a - b)$, we get

Vertical method:

$$\begin{array}{r} \text{LHS} = (a+b)(a-b) \\ a + b \\ \times a - b \\ \hline a^2 + ab \\ - ab \quad \times b^2 \\ \hline a^2 - b^2 \end{array}$$

Horizontal method:

$$\begin{aligned} \text{LHS} &= (a + b)(a - b) = a(a - b) + b(a - b) \\ &= a^2 - ab + ba - b^2 \quad (\text{because } ab = ba) \\ &= a^2 - b^2 = \text{RHS} \quad (\text{and } ab - ab = 0) \end{aligned}$$

Thus LHS = RHS Hence verified.

The above identity can be described in words as ‘**product of the sum and difference of two terms is equal to the difference of their squares.**’

Example 1: Verify $(x + 2)(x - 2) = x^2 - 4$

$$\begin{aligned} \text{Verification: LHS} &= (x + 2)(x - 2) = x^2 - (2)^2 \\ &= x^2 - 4 = \text{RHS} \end{aligned}$$

Thus LHS = RHS Hence verified.

Example 2: Verify $(2x + y)(2x - y) = 4x^2 - y^2$

$$\begin{aligned} \text{Verification: LHS} &= (2x + y)(2x - y) = (2x)^2 - (y)^2 \\ &= 4x^2 - y^2 = \text{RHS} \end{aligned}$$

Thus LHS = RHS Hence verified.

EXERCISE 8.7

I. Verify:

- | | |
|---------------------------------|---------------------------------|
| (1) $(3x+2)(3x-2) = 9x^2 - 4$ | (2) $(4x+3)(4x-3) = 16x^2 - 9$ |
| (3) $(5x+4)(5x-4) = 25x^2 - 16$ | (4) $(3x+7)(3x-7) = 9x^2 - 49$ |
| (5) $(6x+5)(6x-5) = 36x^2 - 25$ | (6) $(7x+6)(7x-6) = 49x^2 - 36$ |

II. Verify the following:

- | | |
|--|---|
| (1) $4a^2 - 9b^2 = (2a + 3b)(2a - 3b)$ | (2) $64m^2 - 81 = (8m + 9)(8m - 9)$ |
| (3) $9p^2 - 100q^2 = (3p + 10q)(3p - 10q)$ | (4) $100y^2 - 169 = (10y + 13)(10y - 13)$ |

8.4 FACTORIZATION OF ALGEBRAIC EXPRESSIONS

8.4.1 Factorize an algebraic expression (using algebraic identities).

The factors of a number, or an expression, can be expressed as the product of given number or given expressions. Such numbers or expressions are called factors of the given number or given expression.

For example: $12 = 2 \times 2 \times 3$, $4a^2 = 2 \times 2 \times a \times a$, $a^2 + 5a + 6 = (a + 3)(a + 2)$

We have learnt to find the factors of numbers but here we shall learn to find the factors of an algebraic expression.

(i) Factors of expression of the form: $ka + kb + kc$

There are three terms in the given expression $ka + kb + kc$ and each term can be divided by k (i.e., the common factor of every term).

$$\text{As } \frac{ka}{k} = a, \quad \frac{kb}{k} = b, \quad \frac{kc}{k} = c,$$

So, multiplying and dividing each term by k , we get:

$ka + kb + kc = k(a + b + c)$, i.e. there are two factors of the given expression, k and $(a + b + c)$.

Example 1: Find the factors of $2a + 4b + 6c$

Solution: 2 is common factor in all the three terms.

$$\text{i.e. } \frac{2a}{2} = a, \quad \frac{4b}{2} = 2b, \quad \frac{6c}{2} = 3c$$

$$\text{Hence, } 2a + 4b + 6c = 2(a + 2b + 3c)$$

Example 2: Find the factors of $12a^2 - 8ab + 4a$

Solution: a , $2a$ and $4a$ are the common factors in the expression but $4a$ is the highest common factor.

$$\text{As } \frac{12a^2}{4a} = 3a, \quad \frac{8ab}{4a} = -2b, \quad \frac{4a}{4a} = 1$$

$$\text{Hence, } 12a^2 - 8ab + 4a = 4a(3a - 2b + 1)$$

Example 3: Find the factors of $a(x + y) + b(x + y) + c(x + y)$

Solution: $(x + y)$ is the common factor of all the three terms in the given expression.

$$\text{i.e. } a(x + y) + b(x + y) + c(x + y) = (x + y)(a + b + c)$$

EXERCISE 8.8

I. Factorize the following.

- | | | |
|---|--|-----------------------------|
| (1) $5a^2x - 15a$ | (2) $m^4 - m^3 + m^2$ | (3) $7a^3 + 14a^2 - 7$ |
| (4) $mx^2 - m^2x - mx$ | (5) $13n - 26n^3 + 39n^5$ | (6) $a^2bc + ab^2c + abc^2$ |
| (7) $3a(b + c) + 6d(b + c)$ | (8) $xy(a + b) + y(a + b)^2 + x(a + b)$ | |
| (9) $4lm(x - y) + 8mn(x - y)^2 + 12nl(x - y)$ | (10) $pq(p^2 - q)^2 + pr(p^2 - q)^2 + qr(p^2 - q)^2$ | |

II. Factors of expression of the form: $(a^2 + 2ab + b^2)$ or $(a^2 - 2ab + b^2)$

We know that $(a + b)^2 = a^2 + 2ab + b^2$

and $(a - b)^2 = a^2 - 2ab + b^2$

The expression $(a^2 + 2ab + b^2)$ or $(a^2 - 2ab + b^2)$ has three terms.

- The first and last terms are perfect squares
- The middle term is twice the product of the square roots of the first

Example 1. Factorize: $a^2 + 6ab + 9b^2$

Solution: In the given expression:

- there are three terms
- the first is the square of a and the last term is the square of $3b$.
- the middle term is according to the condition i.e. $2(a)(3b) = 6ab$.

Therefore the factors of the given expression can be written in the form:

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$\begin{aligned}\text{So, } a^2 + 6ab + 9b^2 &= a^2 + 2(a)(3b) + (3b)^2 \\ &= (a + 3b)^2 \\ &= (a + 3b)(a + 3b)\end{aligned}$$

Example 2. Factorize: $2x^2 + 12x + 18$

Solution: Take the common factor 2 from the given expression and then factorize accordingly.

$$\begin{aligned}\text{i.e. } 2x^2 + 12x + 18 &= 2(x^2 + 6x + 9) && (2 \text{ as common factor}) \\ &= 2\{(x)^2 + 2(x)(3) + (3)^2\} && (\text{perfect square formula}) \\ &= 2(x + 3)^2 \\ &= 2(x + 3)(x + 3)\end{aligned}$$

Example 3. Factorize: $20a^2 - 60a^2b + 45ab^2$

Solution: $20a^2 - 60a^2b + 45ab^2$

$$\begin{aligned}&= 5a(4a^2 - 12ab + 9b^2) && (5a \text{ is common factor}) \\ &= 5a\{(2a)^2 - 2(2a)(3b) + (3b)^2\} && (\text{perfect square formula}) \\ &= 5a(2a - 3b)^2 && (\text{Because } a^2 - 2ab + b^2 = (a - b)^2) \\ &= 5a(2a - 3b)(2a - 3b)\end{aligned}$$

Example 5. Factorize: $98c^2 - 140cd + 50d^2$

Solution: First and last term are not perfect square. If we take 2 as common

$$\begin{aligned} \text{i.e. } 98c^2 - 140cd + 50d^2 &= 2 (49c^2 - 70cd + 25d^2) && \text{(Taking 2 as common)} \\ &= 2 \{(7c)^2 - 2(7c)(5d) + (5d)^2\} \\ &= 2 (7c - 5d)^2 \\ &= 2 (7c - 5d) (7c - 5d) \end{aligned}$$

EXERCISE 8.9

Factorize the following.

- (1) $x^2 + 12x + 36$ (2) $a^2 + 4a + 4$ (3) $4a^2 + 12a + 9$
 (4) $9x^2 + 42xy + 49y^2$ (5) $25a^2 + 80ab + 64b^2$ (6) $16b^2 + 40b + 25$
 (7) $4x^2 + 2x + \frac{1}{4}$ (8) $144x^2 - 4x + \frac{1}{36}$ (9) $18c^2 + 60cd + 50cd^2$
 (10) $4a^2 + 48a + 144$ (11) $5y^2 + 40y + 80$ (12) $3s^2 - 48st + 192t^2$
 (13) $9a^2t^2 - 42ats + 49s^2$ (14) $4a^4 - 12a^2 + 9$ (15) $50p^2q^2 - 70pqr + 98r^2$

III. Factors of an algebraic expression of the form $a^2 - b^2$

We already know that $(a + b)(a - b) = a^2 - b^2$

or vice versa $a^2 - b^2 = (a + b)(a - b)$

The characteristics are

- (1) There are two square terms.
- (2) Negative sign is between the two terms.

Method for finding the factors

- (i) Find the square root of both the terms.
- (ii) Write the two terms in two small brackets in the product form,

Example 1. Factorize: $4x^2 - 9y^2$

Solution: There is difference of two squares

$$\begin{aligned} \text{Hence, } 4x^2 - 9y^2 &= (2x)^2 - (3y)^2 \\ &= (2x + 3y)(2x - 3y) \end{aligned}$$

Example 2. Factorize: $9y^2 - 81$

Solution: Taking first the common factor 9,

$$\begin{aligned} 9y^2 - 81 &= 9(y^2 - 9) \\ &= 9(y + 3)(y - 3) \end{aligned}$$

Example 3. Factorize: $15(a + b)^2 - 60(c + d)^2$

Solution:

$$\begin{aligned} 15(a + b)^2 - 60(c + d)^2 &= 15\{(a + b)^2 - 4(c + d)^2\} \\ &= 15\{(a + b) + 2(c + d)\}\{(a + b) - 2(c + d)\} \\ &= 15(a + b + 2c + 2d)(a + b - 2c - 2d) \end{aligned}$$

EXERCISE 8.10

Factorize the following.

- (1) $b^2 - c^2$ (2) $a^2 - 36$ (3) $a^2 - 49$ (4) $25 - y^2$
 (5) $4x^2 - 9$ (6) $9x^2 - 16y^2$ (7) $6x^2 - 24$ (8) $200 - 18z^2$
 (9) $81a^2 - 121b^2$ (10) $a^2c^2 - 16c^2d^2$ (11) $36 - (3a - 2b)^2$
 (12) $(2x + 3y)^2 - 100z^2$ (13) $64b^2 - (6b + d)^2$ (14) $(3a - 4b)^2 - (3a - b)^2$
 (15) $49(2a - 3b)^2 - 16(a + 2b)^2$ (16) $225(p - q)^2 - 49(p + q)^2$
 (17) $72(c + d)^2 - 162(c - d)^2$ (18) $48l(m - n)^2 - 75l(p - q)^2$

8.4.2 Factorize an algebraic expression making groups.

The method of factorizing an algebraic expression by making groups is explained with the help of the following example.

Example 1. Factorize

(i) $x^2 + 5x + 4x + 20$

Solution:

$$\begin{aligned} &(i) \ x^2 + 5x + 4x + 20 \\ &= (x^2 + 5x) + (4x + 20) \text{ (Making groups)} \\ &= x(x + 5) + 4(x + 5) \\ &= (x + 5)(x + 4) \text{ [Taking } (x + 5) \text{ as common factor]} \end{aligned}$$

(ii) $x^2 + 4y^2 + xy^2 + 4x$

Solution:

$$\begin{aligned} &(ii) \ x^2 + 4y^2 + xy^2 + 4x \\ &= (x^2 + 4x) + (xy^2 + 4y^2) \text{ (By making groups)} \\ &= x(x + 4) + y^2(x + 4) \\ &= (x + 4)(x + y^2) \text{ [Taking } x + 4 \text{ as common factor]} \end{aligned}$$

In case of expression of three terms which is not a complete square, so factorize the last term in such a way that its factors, either added or subtracted, give the coefficient of middle term.

We proceed with such expression as under:

Example 2. Factorize: $x^2 + 5x + 6$

Solution: Factorizing the term 6 in such a way that its two factors when added gives the coefficient of middle term.

$$\text{So, } x^2 + 5x + 6 = x^2 + 3x + 2x + 6$$

It is observed that two factors are 2 and 3 has common variable.

$$= (x^2 + 3x) + (2x + 6)$$

$$= x(x + 3) + 2(x + 3) \quad (\text{Making their groups})$$

$$= (x + 3)(x + 2) \quad [\text{Take } (x + 3) \text{ as common factor}]$$

Example 3. Factorize: $x^2 + 5x - 6$

Solution:

$$x^2 + 5x - 6$$

$$= x + 6x - x - 6 \quad (\text{Factorizing the last term})$$

$$= (x + 6x) + (-x - 6) \quad (\text{Breaking the middle term})$$

$$= x(x + 6) - 1(x + 6) \quad (\text{Making their groups})$$

$$= (x + 6)(x - 1) \quad (\text{Taking } (x + 6) \text{ as common factor})$$

EXERCISE 8.11

- I. Factorize:**
- | | |
|----------------------------------|-----------------------------|
| (1) $5x^2 + 5x + 4x + 4$ | (2) $ax + ay + bx + by$ |
| (3) $b^2 + 4c^2 + bc^2 + 4b$ | (4) $p^2 - 6pq - pq + 6q^2$ |
| (5) $a^2 + 5b^2 + ab^2 + 5a$ | (6) $7x^2 + py + 7x + pyx$ |
| (7) $2c^2d + 4cd^2 - 2cd - 4d^2$ | (8) $2y^2 - 10y + 4y - 20$ |
| (9) $x^2y^2 + 5xy - xy - 5$ | (10) $p^2 - pq + q - p$ |
- II. Factorize:**
- | | |
|----------------------|-----------------------|
| (1) $x^2 + 6x + 8$ | (2) $y^2 + 8y + 12$ |
| (3) $a^2 + 7a + 10$ | (4) $c^2 + 5c + 6$ |
| (5) $d^2 + 6d + 5$ | (6) $p^2 - 4p + 3$ |
| (7) $r^2 + 7r + 12$ | (8) $a^2 + 3a - 10$ |
| (9) $m^2 - 5m + 14$ | (10) $x^2 - 2x - 15$ |
| (11) $y^2 + 7y - 18$ | (12) $x^2 + 10x + 21$ |

REVIEW EXERCISE 8

1. Answer the following questions.

- (i) What is meant by literals? (ii) Define a constant.
(iii) What is a binomial? (iv) What is an algebraic identity?
(v) Define the term factorization of an algebraic expression.

2. Fill in the blanks.

- (i) $(a + b)^2 =$ _____ (ii) $(a - b)^2 =$ _____
(iii) $(x + a)(x + b) =$ _____ (iv) $a - b^2 =$ _____
(v) A symbol represented by a literal and can take various numerical values is called a _____.
(vi) A polynomial having only one term is called _____.

3. Tick (✓) the correct answer.

- (i) $x^2 - x = ?$
(a) x (b) $x(x - 1)$ (c) x^2 (d) $x - x^2$
(ii) A polynomial having two terms is called a:
(a) factorization (b) monomial (c) binomial (d) trinomial
(iii) A symbol having a fixed value is called a:
(a) term (b) variable (c) constant (d) literal
(iv) The factors of $a^2 - 9$ are:
(a) $(a + 3)(a - 3)$ (b) $(a + 9)(a - 9)$
(c) $(a - 3)(a - 3)$ (d) $(a - 9)(a - 9)$
(v) $(x - y)(x - y) = ?$
(a) $x^2 - y^2$ (b) $x^2 + 2xy + y^2$
(c) $x^2 - 2xy + y^2$ (d) $x^2 + y^2$

4. If $A = 2(x^2 + y^2 + z^2)$, $B = -x^2 + 3y^2 + 2z^2$ and $C = x^2 - y^2 - 3z^2$, then find:

- (i) $A + B + C$ (ii) $B + C - A$ (iii) $A - B + C$
(iv) $A + B - C$ (v) $A - B - C$ (vi) $B - C - A$

5. Simplify the following polynomials.

- (i) $(x - 2y)(x + 2y)$ (ii) $(4x^2)(3x + 1)$
 (iii) $2x(x + y) - 2y(x - y)$ (iv) $(a^2b^3)(2a - 3b)$
 (v) $(a^2 - b^2)(a^2 + b^2)$ (vi) $(a^2 + 1)(a^2 - a - 1)$
 (vii) $x(y + 1) - y(x + 1) - (x - y)$
 (viii) $a^2(b^2 - c^2) + b^2(c^2 - a^2) + c^2(a^2 - b^2)$

6. Verify the following.

- (i) $(3x - 5)(3x + 5) = 9x^2 - 25$ (ii) $(2a - 5b)^2 = 4a^2 - 20ab + 25b^2$
 (iii) $(3x^2 + 4y^2)^2 = 9x^4 + 24x^2y^2 + 16y^4$ (iv) $9x^2 - 25y^2 = (3x - 5y)(3x + 5y)$

7. Resolve into factors.

- (i) $10a^2 - 200a^4b$ (ii) $36x^3y^3z^3 - 27x^2y^4z + 63xyz^4$
 (iii) $15x^4y + 21x^3y^2 - 27x^2y^2 - 33xy^4$ (iv) $x(a^2 + 11) - 16(a^2 + 11)$
 (v) $x^2(ab + c) + xy(ab + c) + y^2(ab + c)$

8. Factorize the following.

- (i) $a^2 - 26a + 169$ (ii) $1 - 6x^2y^2z + 9x^4y^4z^2$ (iii) $7ab^2 - 343a$
 (iv) $75 - 3(x - y)^2$ (v) $49(x + y)^2 - 16(x - y)^2$ (vi) $\frac{9}{16}a + ab + \frac{4}{9}b^2$
 (vii) $\frac{a^2}{b^2}l^2 - \frac{2ac}{bd}lm + \frac{c^2}{d^2}m^2$ (viii) $(a - \frac{9}{5})^2 - \frac{36}{25}m^2$

SUMMARY

- ➔ Algebra is used to create mathematical models of real world situation.
- ➔ The letters or alphabets that we use to represent unknowns /
- ➔ A symbol represented by a literal and can take various numerical
- ➔ A symbol having a fixed value is called a constant.
- ➔ A combination of constants and variables connected by the signs

- ➔ The parts of an algebraic expression separated by the operational
- ➔ Co-efficient is the number multiplied by a variable or variables.
- ➔ If the sum of the co-efficients is zero then the whole term becomes zero.
- ➔ An algebraic expression in which the exponents of variables are
- ➔ An expression having one term is called monomial.
- ➔ An expression having two terms is called binomial.
- ➔ An expression having three terms is called trinomial.
- ➔ An algebraic expression can be factorized by making groups.
- ➔ A polynomial can be arranged in any order but usually we arrange
- ➔ An algebraic equation which is true for all values of the variable
- ➔ The process of writing an algebraic expression as the product of two or more expressions which divide it exactly is called the

$$(i) \quad (a + b)^2 = (a + b)(a + b) = a^2 + 2ab + b^2$$

$$(ii) \quad (a - b)^2 = (a - b)(a - b) = a^2 - 2ab + b^2$$

$$(iii) \quad a^2 - b^2 = (a - b)(a + b)$$

...

Think Tank: The Most Wonderful Knowledge.

6 is a perfect number because $6 = 1 \times 2 \times 3 = 1 + 2 + 3 = 6$

28 is a perfect number because its divisors are:

1, 2, 4, 7, 14 and $1 + 2 + 4 + 7 + 14 = 28$

Find three more perfect numbers.

9.1 LINEAR EQUATIONS

9.1.1 Define a Linear equation in one variable

Equations play an important role in solving daily life problems in algebra. In previous class we have learnt, how to construct an equation.

For example, $2x + 3 = 11$ is an equation. Here x is used as variable, therefore it is an equation in one variable. Since in this equation the exponent of x is 1. So, it is an equation of one degree or first degree equation. The equation which contains a single variable with degree 1 is called linear equation in one variable.

Activity: Tick (✓) linear equations in one variable.

$y^2 + 3$	$4x + 5 = 2$	$x^2 + 3x + 4 = 0$
$3x^2 + 7 = 0$	$x + 5 = 20$	$x + x = 20$
$x^2 + y^2 = 9$	$m = n + 5$	$x + 7 = 11y$
$7v - 3 = 4$	$a^2 + 2ab + b^2 = 0$	$x = 13 - 4$

9.2 SOLUTION OF LINEAR EQUATIONS

9.2.1 Demonstrate different techniques to solve linear equation.

I. Solving simple linear equation by isolating the variable.

The important thing to remember about any equation is that the sign of equality which represents a balance. So, if we do operation to one side of the equation, we have to do it to the other side of the equation. So long as it's done **exactly** the same way on each side, the balance will be preserved.

Our first step in solving any equation is to gather all the terms involving variables and the constant terms separately.

We do all this by using properties of equation.

Example 1. Solve: $3x + 15 = x + 25$

Solution: $3x + 15 = x + 25$

$$\text{or } 3x - x + 15 = x - x + 25 \text{ (Subtracting } x \text{ from each side)}$$

$$\text{or } 2x + 15 = 25$$

$$2x + 15 - 15 = 25 - 15 \text{ (Subtracting 15 from both sides)}$$